PHARYNX AND LARYNX

BY

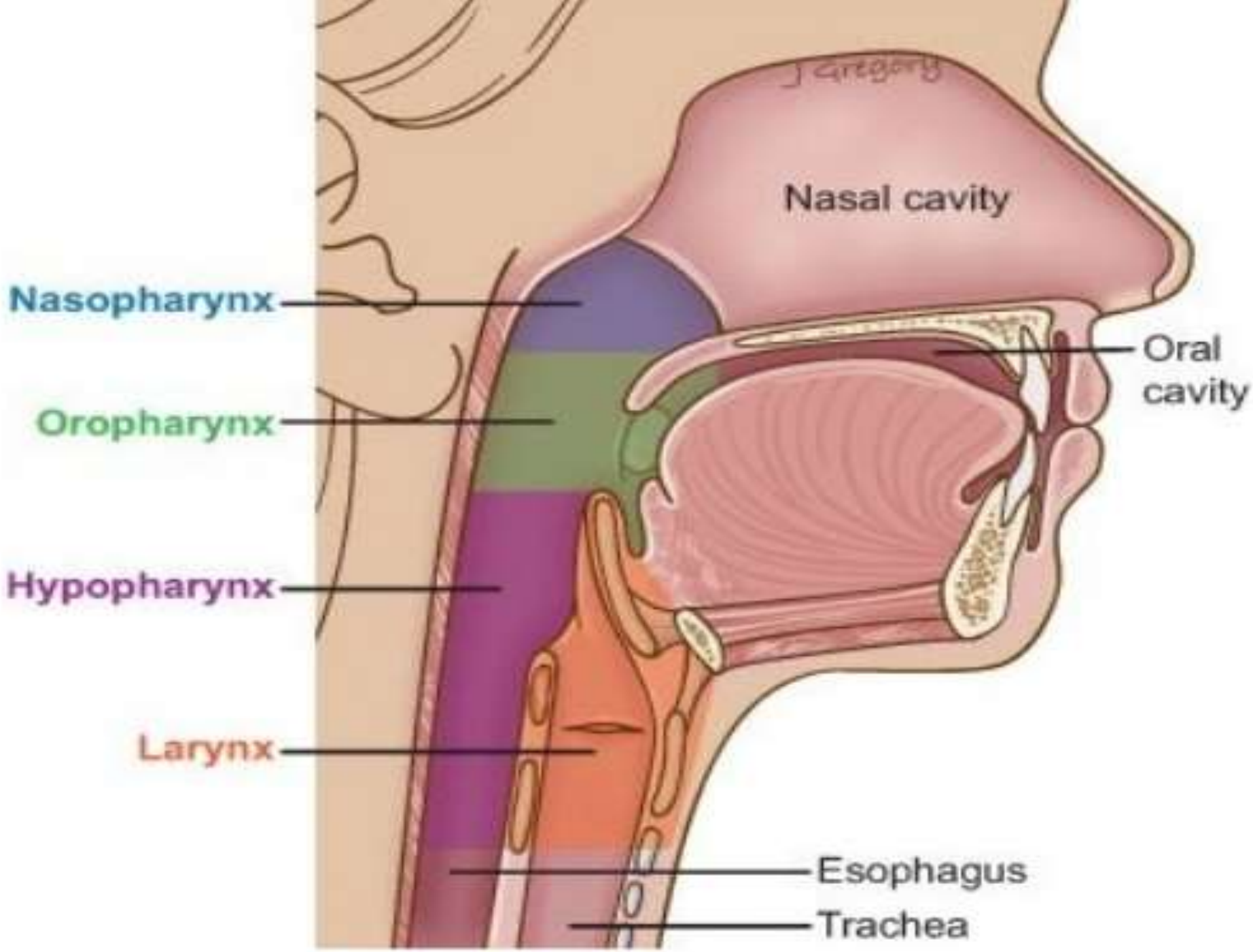
F.O. OPOOLA

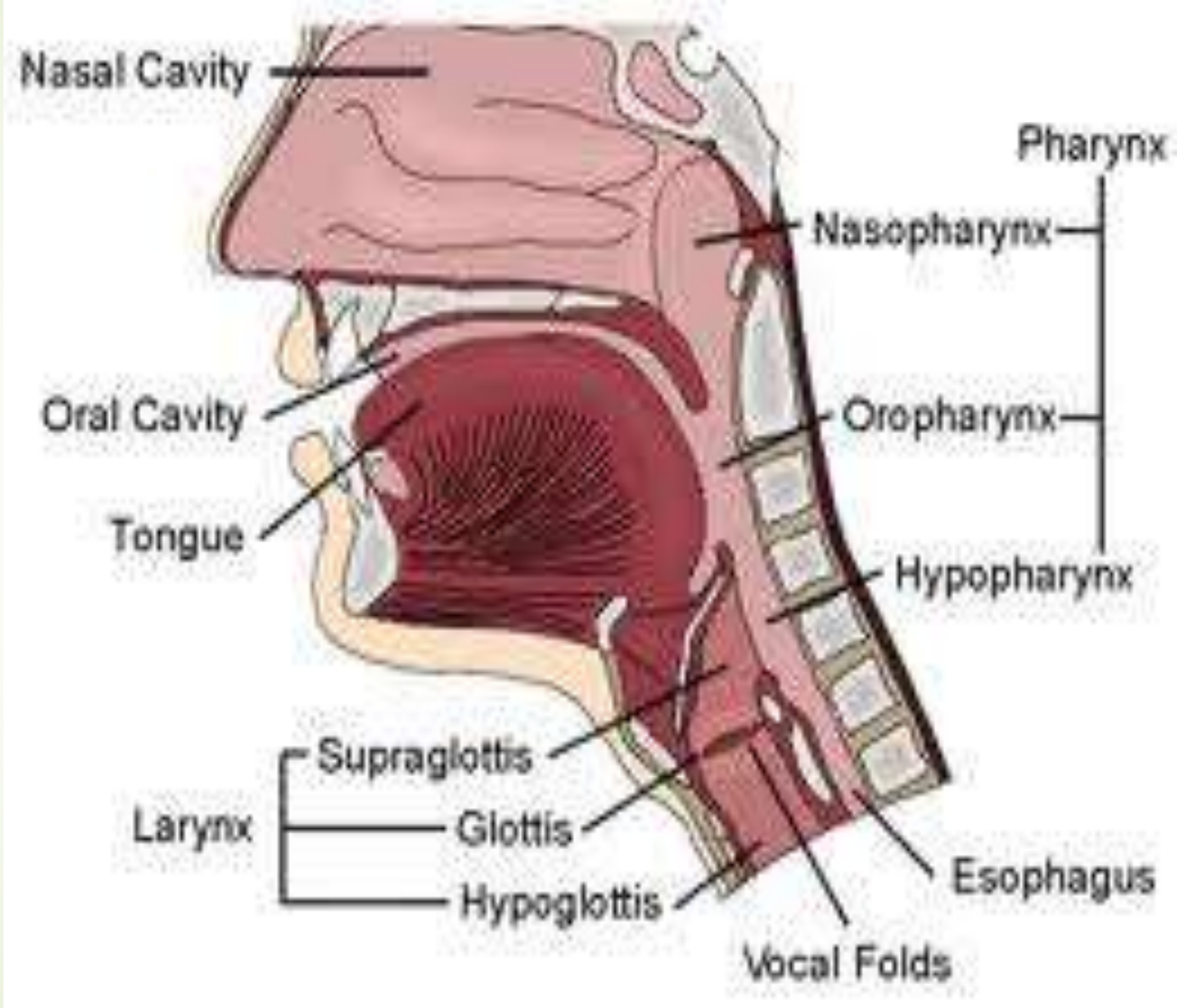
LEARNING OBJECTIVES

- Introduction
- **Pharynx**

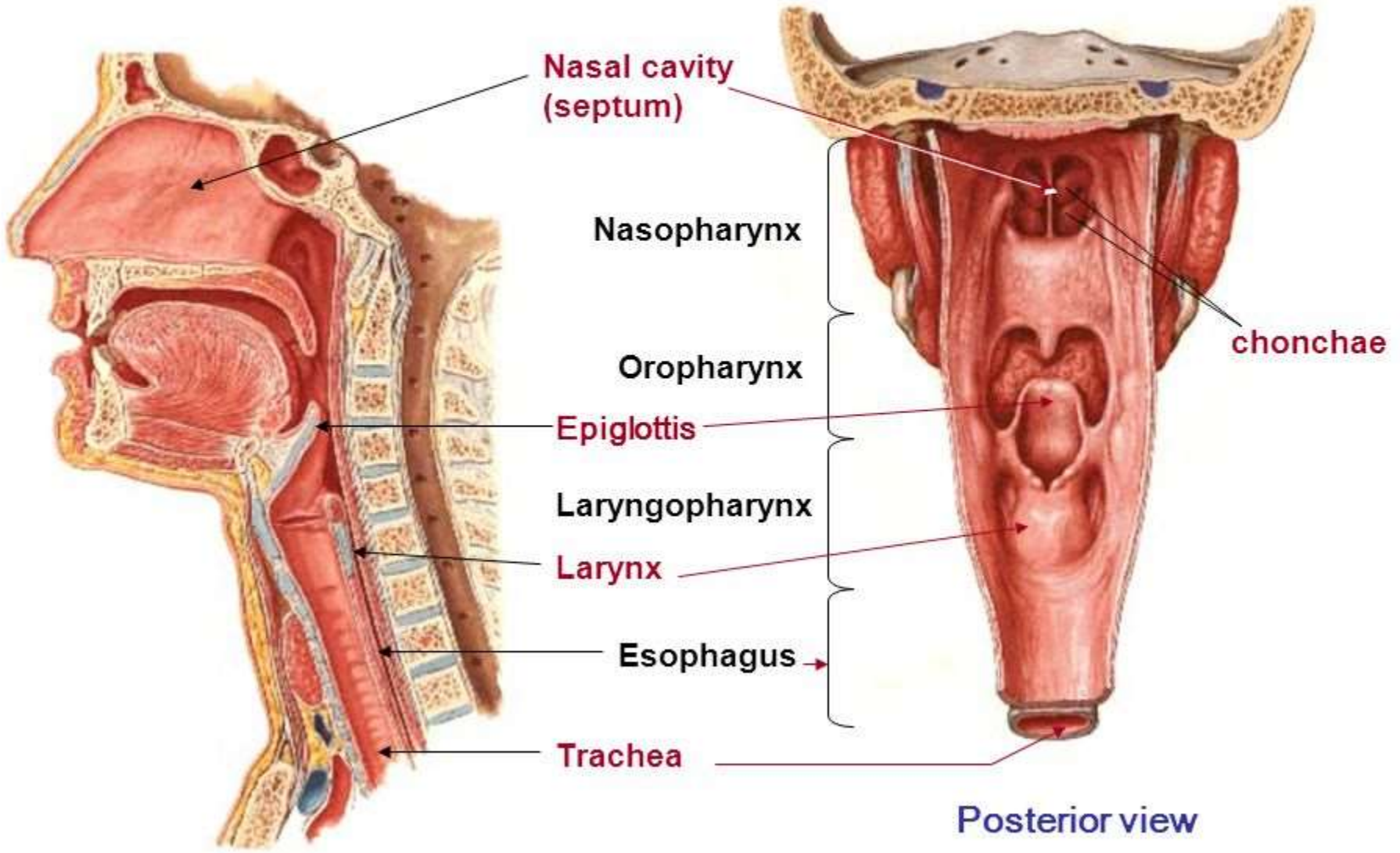
  -Subdivisions and borders

- Pharyngeal muscles
- Blood supply and innervation
- **Larynx**
- Laryngeal cartilages
- Membranes and ligaments
- Vocal cords and folds Muscles and innervation
- Blood supply


Nasal cavity
Nasopharynx
Oropharynx
Hypopharynx
Larynx
Oral cavity
Esophagus
Trachea


Nasal Cavity
Pharynx
Nasopharynx
Oral Cavity
Oropharynx
Tongue
Hypopharynx
Supraglottis
Larynx
Glottis
Hypoglottis
Esophagus
Vocal Folds

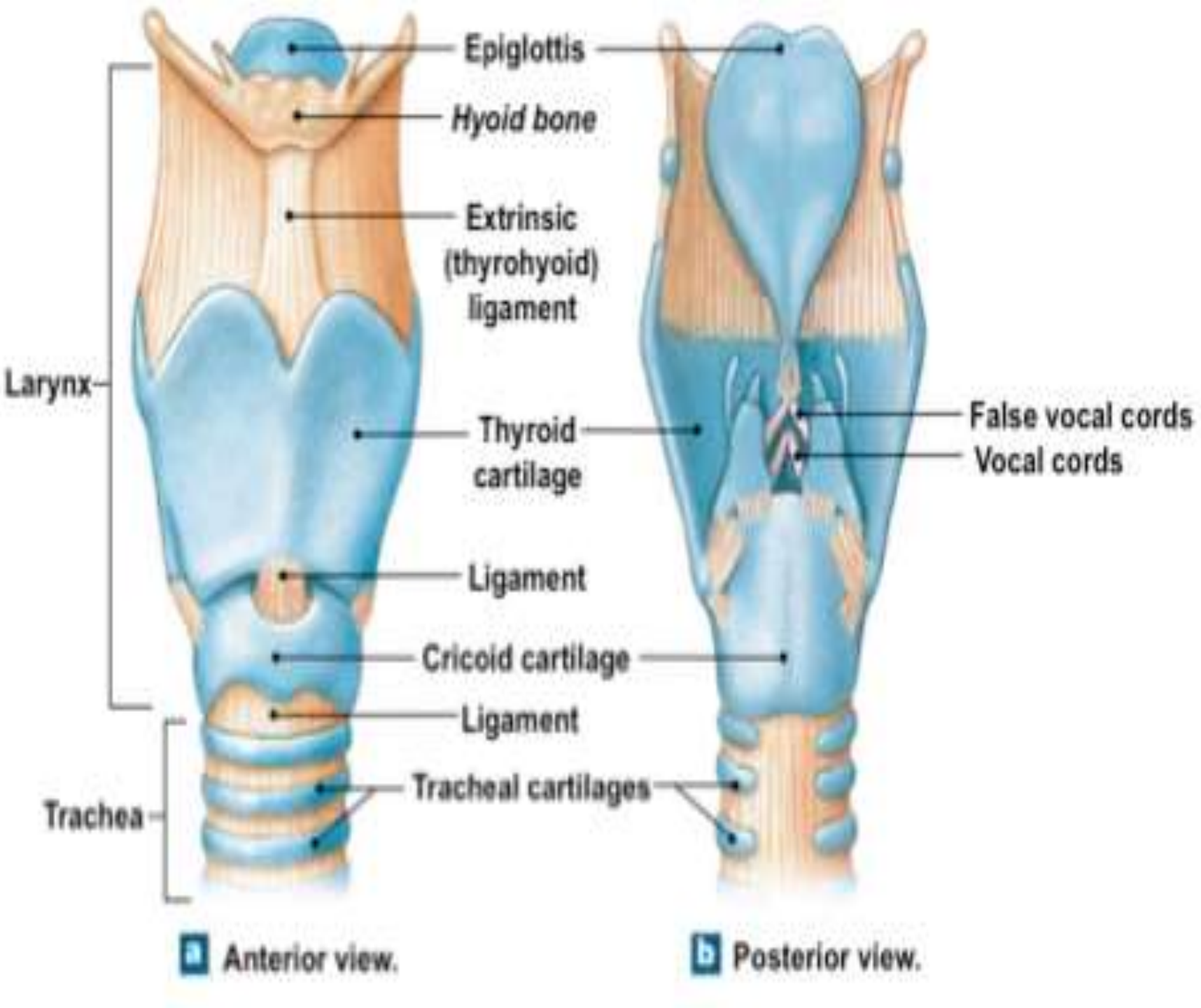
Nasal cavity, Pharynx and Larynx



LARYNX

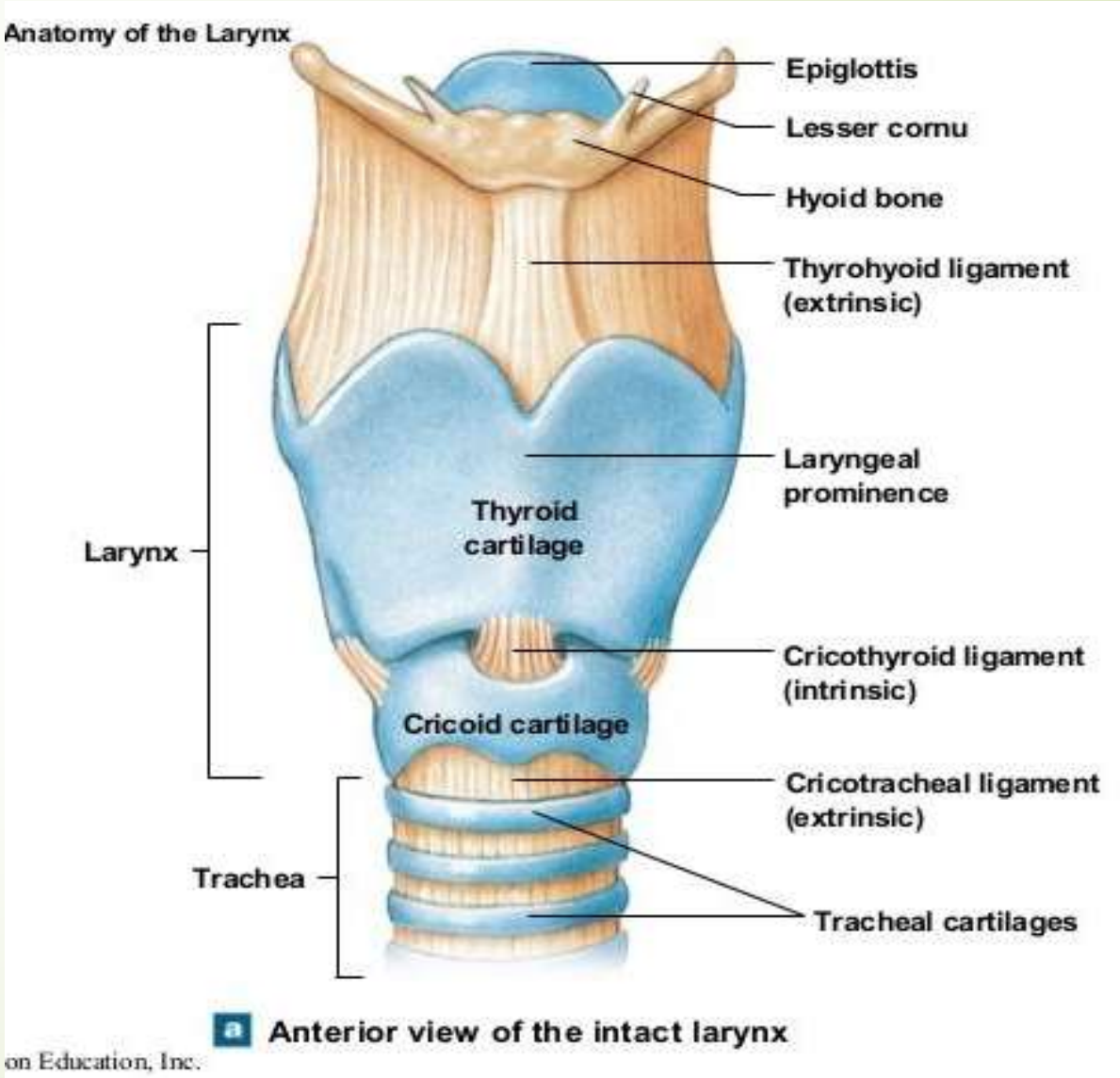
- Is the upper , expanded part of the windpipe which is modified for phonation

- Supported by a number of cartilages



LARYNX

- Complex organ of voice production
- Composed of 9 cartilages connected by membranes and ligaments
- Contains the vocal folds
- Located anteriorly in the neck
- Vertebral levels C3-C6
- Connects the inferior part of the Oropharynx to the trachea
- Lies anterior to the Laryngopharnx
- Secondary function is to guard the air passages



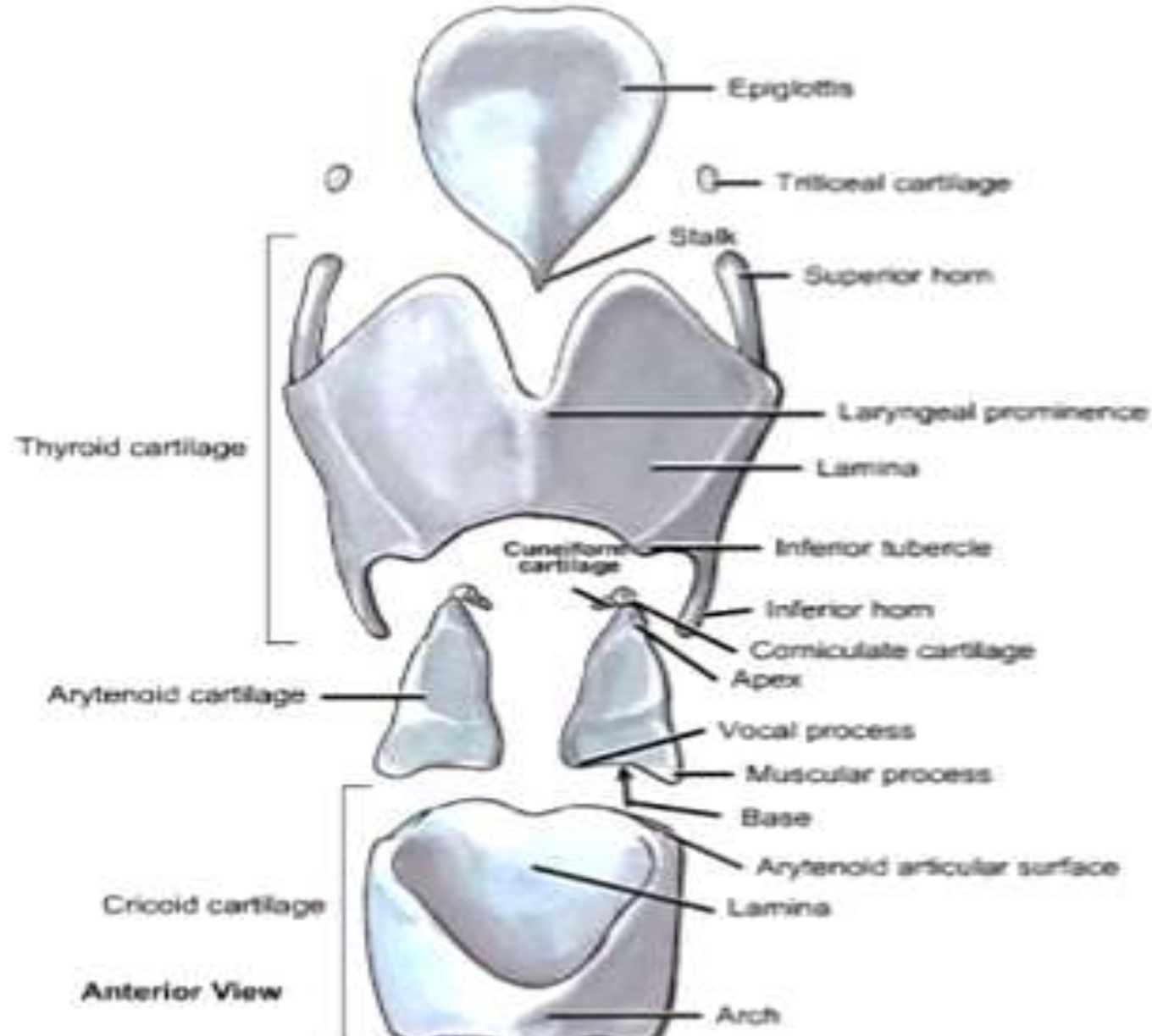
Cartilages

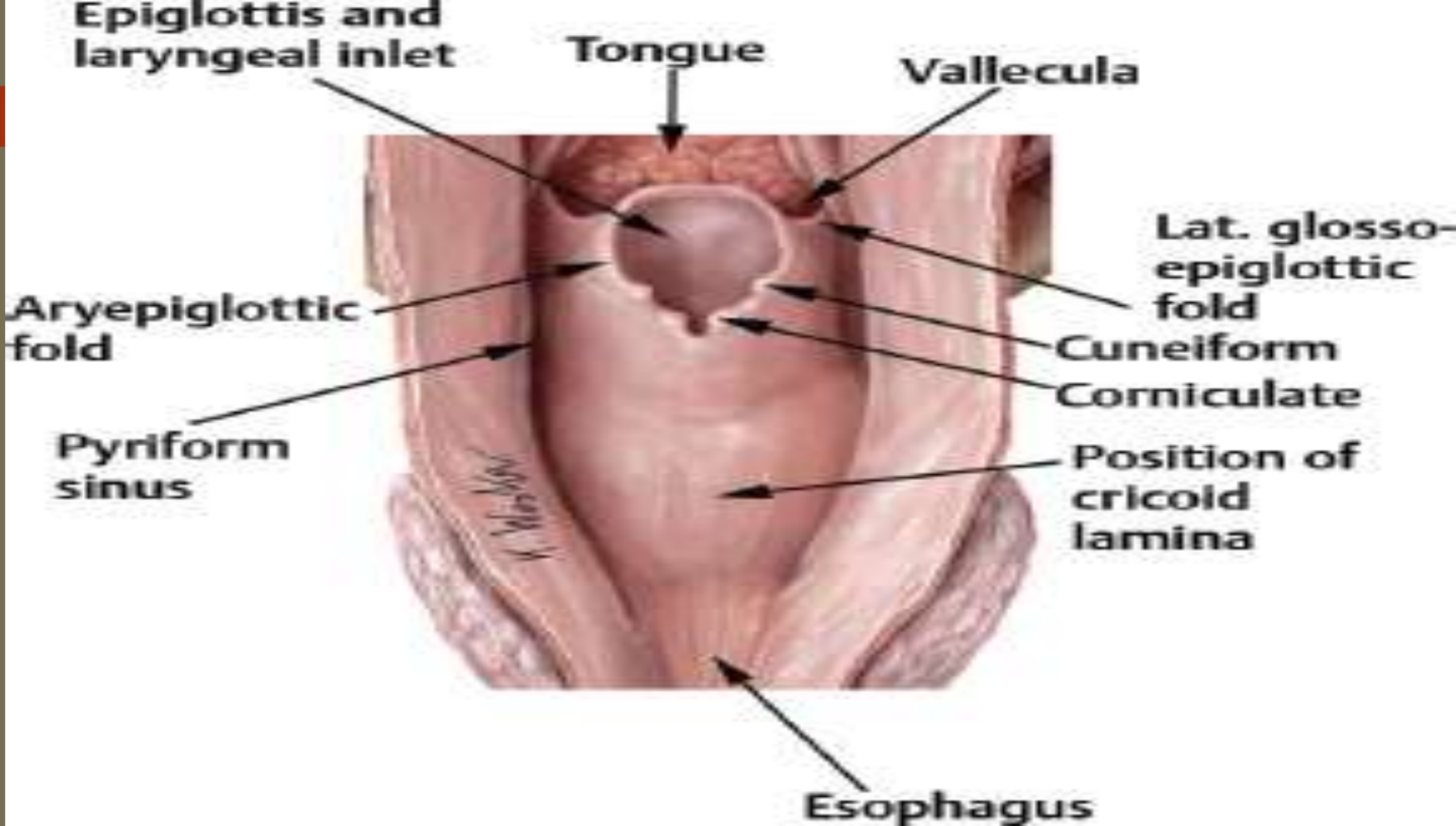
3 Unpaired

- Thyroid
- Cricoid
- Epiglottis

3 Paired

- Arytenoid
- Corniculate
- Cuneiform




Epiglottis and laryngeal inlet
Tongue
Vallecula
Lat. glosso-epiglottic fold
Aryepiglottic fold
Cuneiform
Corniculate
Pyriform sinus
Position of cricoid lamina
Esophagus

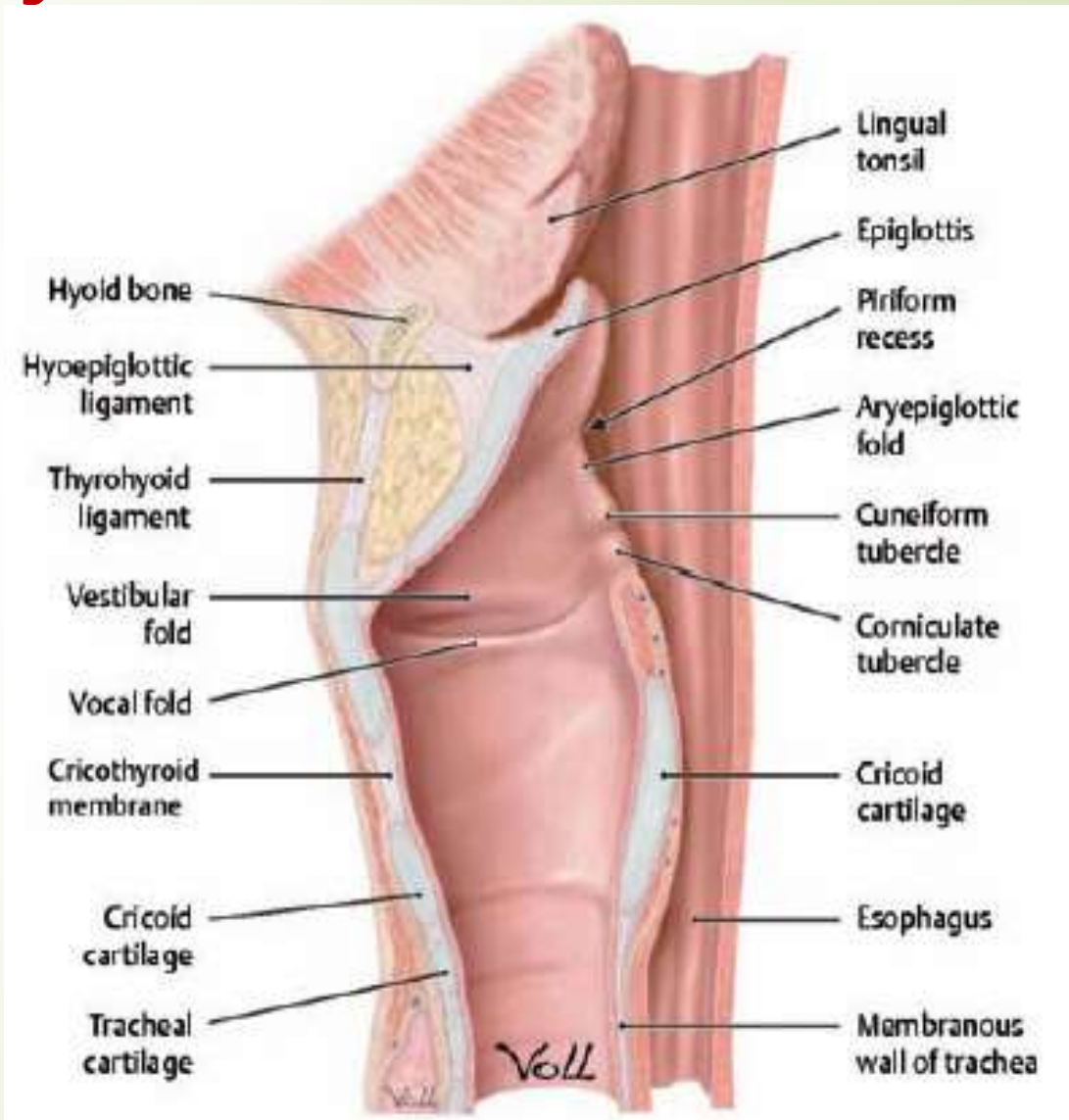
Interior of the Larynx

- Main divisions of the Laryngeal Cavity

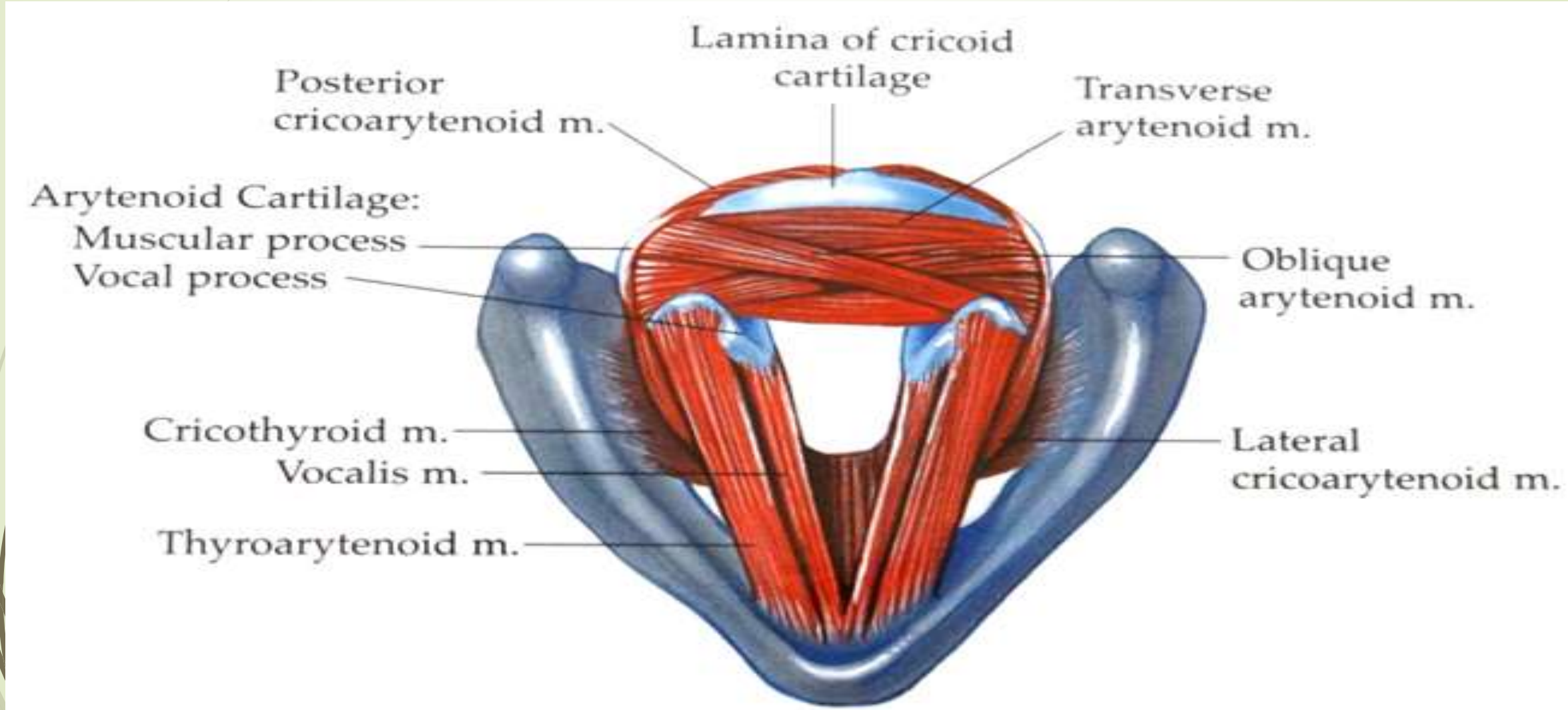
1- Laryngeal Vestibule

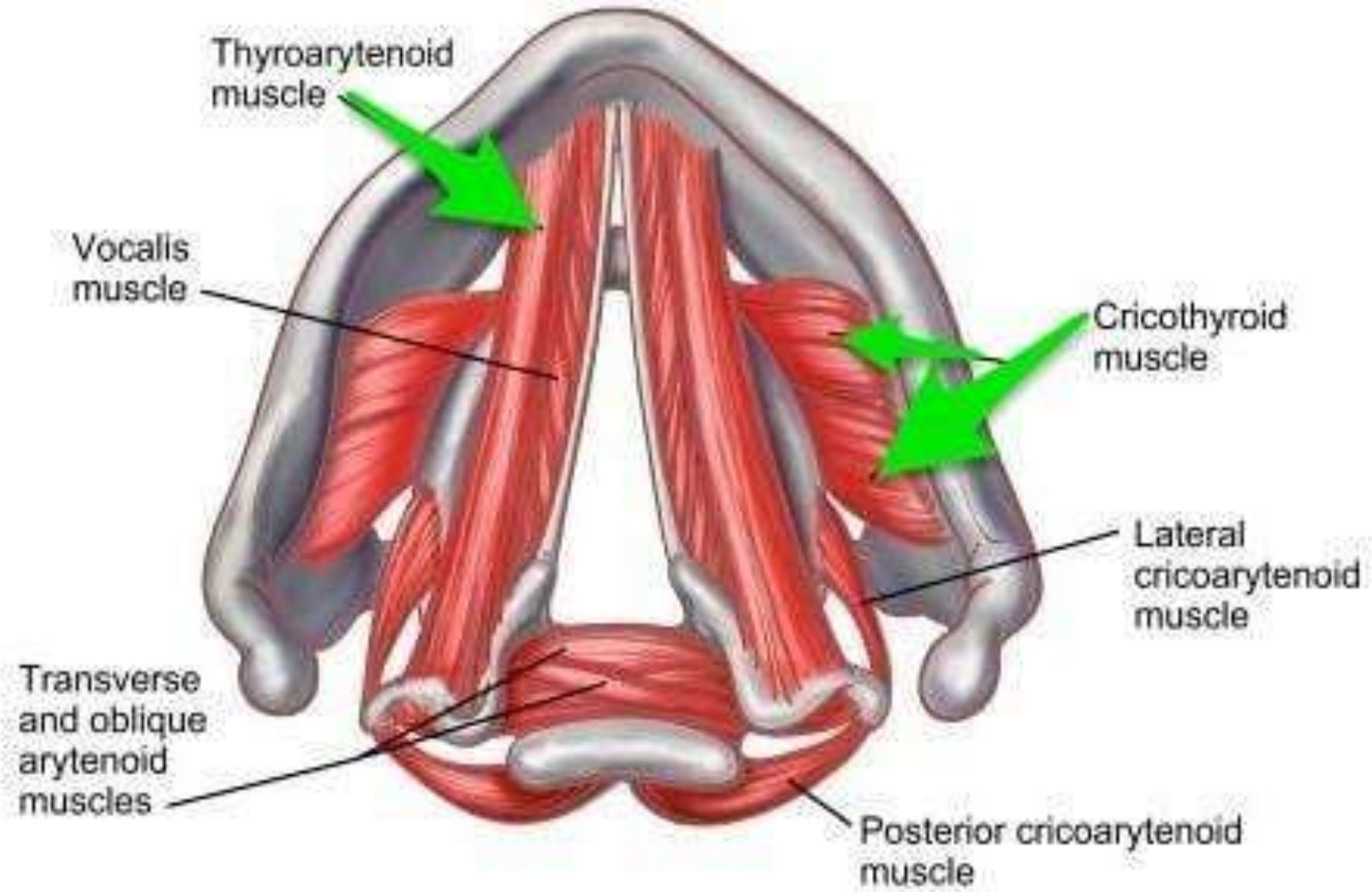
2- Laryngeal cavity

3- Infraglottic cavity



Muscles of the Larynx




Thyroarytenoid muscle
Vocalis muscle
Cricothyroid muscle
Lateral cricoarytenoid muscle
Transverse and oblique arytenoid muscles
Posterior cricoarytenoid muscle

NERVE SUPPLY OF LARYNX

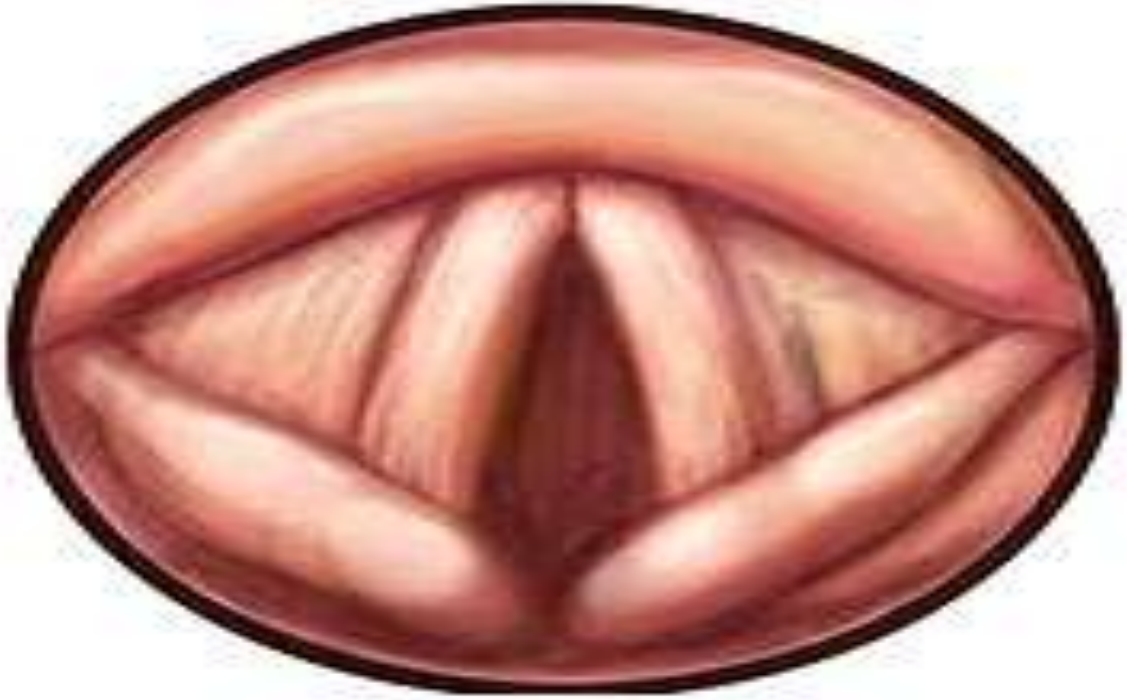
MOTOR

- All the muscles which move the vocal cords (abductors, adductors or tensors) are supplied by the ***Recurrent Laryngeal Nerve*** except the cricothyroid muscle, which is supplied by ***Superior Laryngeal Nerve***.

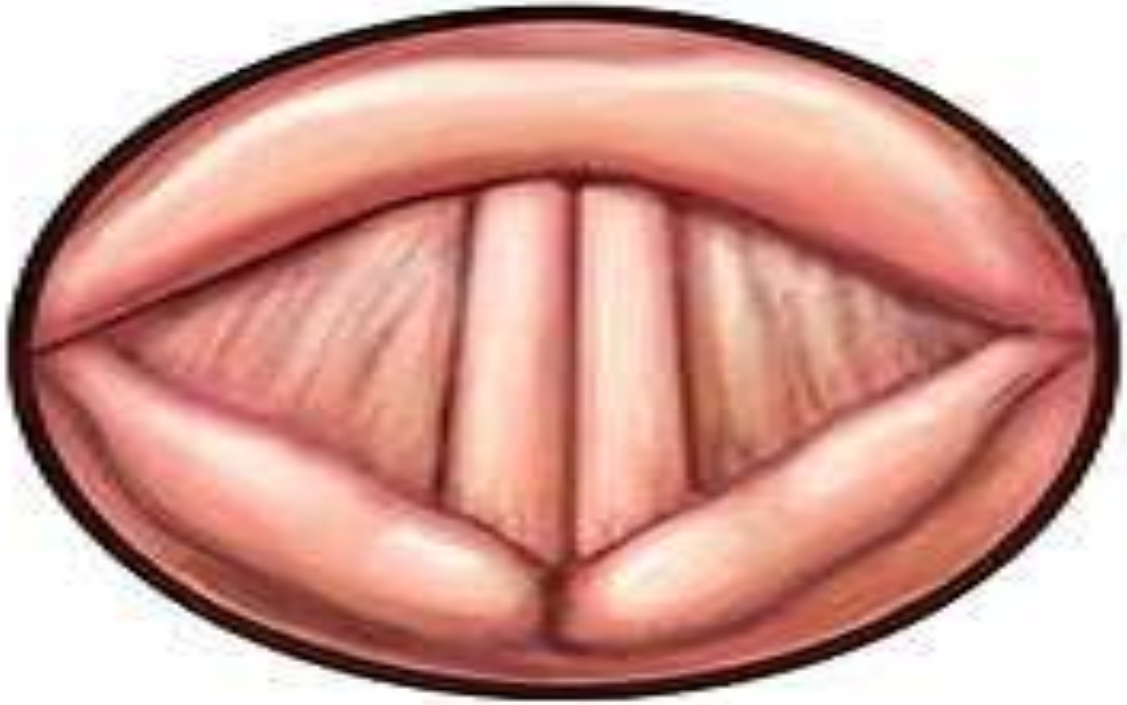
- Both of these are branches of the **Vagus Nerve**.

SENSORY

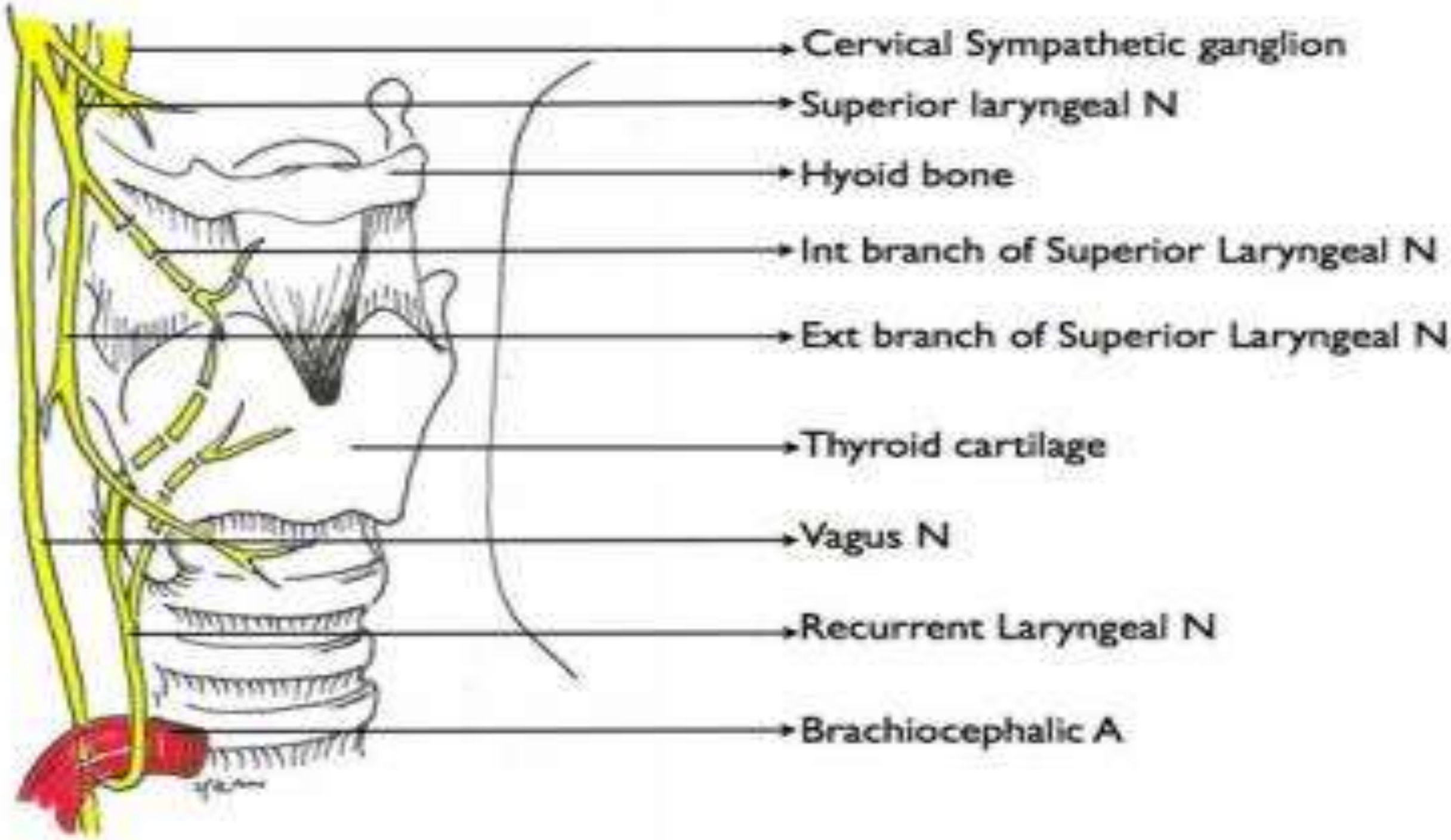
- Above the vocal cords, larynx is supplied by ***Internal Laryngeal Nerve*** – a branch of Superior Laryngeal Nerve & below the vocal cords by ***Recurrent Laryngeal Nerve***.



Vocal cords open during breathing to allow air into lungs.



Vocal cords close when speaking so air from the lungs presses between them to cause the vibrations that produce sound.


Cervical Sympathetic ganglion
Superior laryngeal N
Hyoid bone
Int branch of Superior Laryngeal N
Ext branch of Superior Laryngeal N
Thyroid cartilage
Vagus N
Recurrent Laryngeal N
Brachiocephalic A

Muscle	Origin	Insertion	Innervation	Action
Cricothyroid	Antero-lateral part of cricoid cartilage	Inferior margin and inferior horn of thyroid cartilage	External laryngeal nerve	Stretches and tenses vocal ligament
Thyro-arytenoid	Lower ½ of posterior aspect of angle of thyroid laminae and cricothyroid ligament	Anterolateral arytenoid surface	Inferior laryngeal nerve (terminal part of recurrent laryngeal nerve from vagus)	Relaxes valocal ligament
Posterior Crico-arytenoid	Posterior surface of lamina of cricoid cartilage	Vocal process of arytenoid cartilage		Abducts vocal folds
Lateral crico-arytenoid	Arch of arytenoid cartilage			Adducts vocal folds
Transverse & Oblique arytenoids	One arytenoid cartilage	Contra-lateral arytenoid cartilage		Adducts arytenoid cartilages
Vocalis	Lateral surface of vocal process of arytenoid	Ipsilateral vocal ligament		Relaxes posterior vocal ligament while maintaining tension of anterior part

Blood supply to the Larynx

- **Superior Laryngeal artery**

Supplies the internal surface of the larynx

- **Cricothyroid artery**

Supplies the cricothyroid muscle

- **Inferior Laryngeal artery**

- Supplies the mucous membrane and the muscles of the inferior part of the larynx